

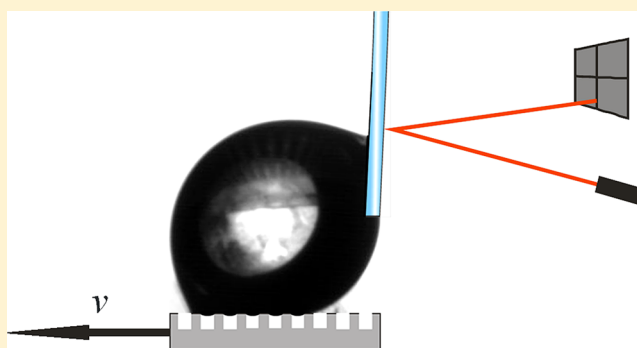
Dynamic Measurement of the Force Required to Move a Liquid Drop on a Solid Surface

D. W. Pilat, P. Papadopoulos, D. Schäffel, D. Vollmer, R. Berger, and H.-J. Butt*

Max Planck Institute for Polymer Research, Ackermannweg 10, 55128 Mainz, Germany

S Supporting Information

ABSTRACT: We measured the forces required to slide sessile drops over surfaces. The forces were measured by means of a vertical deflectable capillary stuck in the drop. The drop adhesion force instrument (DAFI) allowed the investigation of the dynamic lateral adhesion force of water drops of 0.1 to 2 μL volume at defined velocities. On flat PDMS surfaces, the dynamic lateral adhesion force increases linearly with the diameter of the contact area of the solid–liquid interface and linearly with the sliding velocity. The movement of the drop relative to the surfaces enabled us to resolve the pinning of the three-phase contact line to individual defects. We further investigated a 3D superhydrophobic pillar array. The depinning of the receding part of the rim of the drop occurred almost simultaneously from four to five pillars, giving rise to peaks in the lateral adhesion force.



I. INTRODUCTION

Sessile liquid drops on solid surfaces experience lateral adhesion owing to surface tension and contact angle hysteresis. The force needed to overcome lateral adhesion is commonly referred to as hysteresis force, (lateral) capillary force, (lateral) adhesion force, or simply retention or retarding force.^{1–10} Investigating lateral adhesion forces, F_{LA} , on drops is of interest for fundamental wetting science and industrial applications such as self-cleaning surfaces or paint spraying.^{1,3,8,11} In the case of self-cleaning surfaces, drops should slide off easily. Thus, weak lateral adhesion is desired. In the case of paint spraying, fast-moving paint drops need to be slowed down, enabling them to stick to an object. Thus, strong lateral adhesion is desired. Lateral adhesion forces can be distinguished as static and dynamic, depending on whether the drop is stationary or moves relative to the surface. To move liquid drops off a surface, drop movement must first be initiated, which means that a force equal to or greater than the static lateral adhesion force must act on the drop. Then the drop needs to be kept in motion at a defined velocity until the drop leaves the surface. To keep the drop moving, a force equal to the dynamic lateral adhesion force must act on the drop.

The most common method of investigating static lateral adhesion forces is the tilted plate method.^{1,3,7–10,12,13} A drop is placed on a planar horizontal surface. Then the plane is gradually tilted until the drop slides downhill. The sliding or roll-off angle is measured and is related to the projection of the gravitational force needed to initiate drop movement. With the tilted plate method only sufficiently large drops can be investigated (i.e., $V > F_{\text{LA}}/(g\rho_l)$, where V and ρ_l are the volume and mass density of the drop, respectively, and g is the

acceleration of free fall). Small drops do not slide because gravity is not sufficient. The static lateral adhesion forces of small drops can be measured using a centrifuge in which gravitation is replaced by a centrifugal force.^{6,9} The relation between the lateral adhesion force and the shape of the liquid drop was investigated via the tilted plate method.^{1,3,8,13} The lateral adhesion force is proportional to the surface tension, the length of the three-phase contact line (TPCL), and the difference in the cosines of the contact angles at the leading and trailing edges.^{8,13} ElSherbini and Jacobi derived a heuristic formula for the lateral adhesion force by approximating the TPCL with a single ellipse and using an experimentally verified polynomial function for the dependence of the contact angle on the position along the TPCL.¹ For a drop on the verge of sliding on an isotropic and homogeneous surface, they derived

$$F_{\text{LA}} = \frac{24}{\pi^3} \gamma D (\cos \theta_{\min} - \cos \theta_{\max}) \quad (1)$$

Here, γ is the surface tension and D is the diameter of a circle with the same area as the ellipse used to approximate the shape of the TPCL. $\theta_{\min}$ and $\theta_{\max}$ are the maximum and minimum contact angles. The maximum and minimum contact angles are measured at the leading and trailing edges of the drop when the sliding angle is reached.³ We distinguish between maximum/minimum and advancing/receding contact angles because maximum/minimum contact angles are in general not equal to the advancing/receding contact angles.¹⁴ Recent findings

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show that, depending on the initial shape of the drop on the plane, either the maximum contact angle corresponds to the advancing contact angle or vice versa.¹⁵ With the present methods, it is either impossible or cumbersome to gain information about the dynamic lateral adhesion force. Furthermore, it is not sufficient to consider only the static lateral adhesion force because it gives no information about the dynamic lateral adhesion force.¹⁰ To the best of our knowledge, only a few reports exist on the sliding behavior of liquid drops on solid surfaces.^{2,10}

The objective of our work was to measure the dynamic lateral adhesion force $F_{LA}(v)$ of water drops at constant velocities v on different surfaces directly. Our concept of measuring lateral adhesion forces is based on measuring the deflection of a capillary that is stuck in a liquid drop that is sitting on a surface. When the surface moves, the liquid drop pins to the capillary while it is dragged over the surface. Consequently, the deflection of the capillary is a measure of the lateral adhesion forces (Figure 1). In particular, we were able to gain a deeper understanding of the lateral adhesion force experienced by liquid drops when moving over superhydrophobic surfaces.

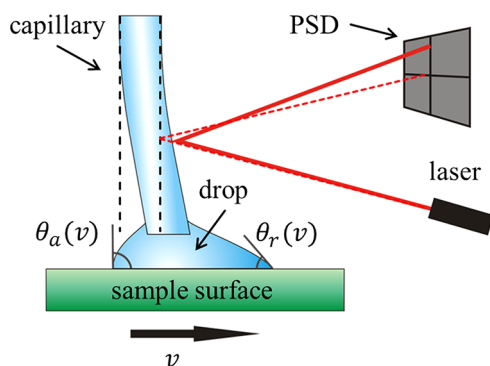


Figure 1. Principle of the drop adhesion force instrument (DAFI): a liquid drop sitting on a solid surface with a rectangular capillary stuck in the drop. The surface underneath the drop is moved at constant velocity v . The drop pins to the capillary and is dragged over the surface. To measure the deflection of the capillary, a laser deflection system (LDS) including a position-sensitive detector (PSD) is used. $\theta_a(v)$ and $\theta_r(v)$ are the dynamic advancing and receding contact angles. If the adhesion between the drop and capillary is larger than the lateral adhesion between the drop and solid surface, then the drop will be dragged over the surface by the capillary.

II. EXPERIMENTAL SECTION

A. Drop Adhesion Force Instrument (DAFI). The DAFI consists of a motorized disk, a fixture holding a capillary, a laser (Schäfer-Kirchhoff, 676 nm $\leq$ 3.7 mW), a position-sensitive detector (PSD, Laser Components 2L20 CP7), two cameras, and a data acquisition system (Figure 2). The capillary, the laser, and the PSD compose a position measurement system based on the triangulation principle (i.e., a laser deflection system (LDS)). The distance between the capillary and the PSD is ~ 10 cm. With the capillary in a relaxed state, the incident and emergent angles are $\sim 15^\circ$. The deflection of the capillary was measured in the range of 0 to 0.6 mm using the LDS. The relative movement of the capillary and the solid surface was realized by the rotation of the motorized disk at constant velocity. Two projections of the drop shape, one on the y - z plane (parallel) and one on the x - z plane (orthogonal), were recorded using the cameras.

A flat or microstructured solid sample was placed on the motorized disk. On this sample we deposited a drop of water (Sartorius Arium

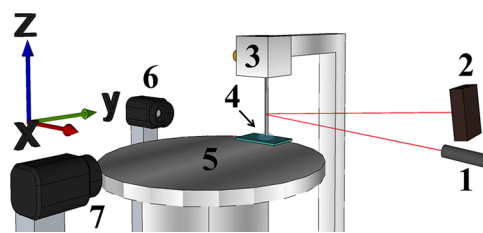


Figure 2. Setup of the DAFI: (1) Laser. (2) Position-sensitive detector, PSD. (3) Fixture with capillary. (4) Liquid drop on a solid sample. (5) Motorized disk. Two cameras are used to monitor the projection of the shape of the drop (6) parallel and (7) orthogonal to the direction of drop movement.

661 VF Water Purification System, specific resistivity of 18.2 M Ω cm) of volume 1 to 2 μ L using a pipet (Thermo Scientific, Finnpiette F2). The sample with the drop was placed next to the capillary. The sample was moved by the motorized disk so that the capillary was in contact with the drop. Then the drop was dragged over the solid sample. The force needed to move the drop (i.e., the dynamic lateral adhesion force) was derived using Hooke's law, $F = ks$, where k is the spring constant of the capillary, F is the force acting on the capillary in the direction of the deflection, and s is the lateral deflection of the capillary. A similar concept was applied in the past by scientists who used glass microneedles to study the driving force on drops on surfaces with wetting gradients.^{16,17} However, using this concept, lateral adhesion forces could not be measured.

1. Calibration of the Glass Capillary Beam. For most experiments, we used a rectangular borosilicate glass capillary (VitroTubes, VitroCom) with a length of 50 mm, a width of 0.36 mm, and a thickness of 0.09 mm. The rectangular shape of the capillary provides a large flat surface with its normal in the x direction for the laser to reflect from while having a smaller spring constant than a quadratic capillary with the same width. A larger area in the direction of drop movement gives rise to a stronger adhesion between the drop and the capillary. To increase the reflectivity of the glass capillary, its broad side was coated with 5 nm chromium and 50 nm gold layers by sputtering (BAL-TEC MED 020). The spring constant k of the capillary in the direction of the normal of the coated surface (i.e., x direction, Figure 2) is given by

$$k = \frac{E(d_o^3 b_o - d_i^3 b_i)}{4L^3} \quad (2)$$

The equation is derived from two components given by the spring constant of a filled beam ($d_o = 0.09$ mm, $b_o = 0.36$ mm, $L = 30$ mm) minus the spring constant of an imaginary inner beam ($d_i = 0.03$ mm, $b_i = 0.3$ mm, $L = 30$ mm). L is the length from the free end of the capillary to the part of the capillary that is fixed by the capillary fixture. This simple approach is possible because both components share the same neutral axis. The thickness of the reflective coating is about 55 nm (i.e., a factor of 10^4 thinner than the capillary dimensions) and can be neglected. Assuming that the nominal dimensions of the capillary and the Young's modulus for borosilicate glass ($E = 63$ GPa, Duran borosilica glass 3.3) are exact gives $k_{\text{nominal}} = 0.148$ N/m. The value of E is only a mean value for borosilicate glass and might differ for the glass used for our capillaries. Furthermore, we made the following two assumptions: First, we assumed that the profile of the capillary is constant along the z direction (Figure 2) (i.e., z -dependent variations of the second moment of area are small). Second, we assumed that the capillary is fixed in the fixture at the point where the capillary comes out of the fixture. Because of thickness variations in the fixture plates squeezing the capillary, the actual point of fixation might be located somewhere inside the fixture, thus increasing the actual L . A deviation of L of about 5 mm can cause a deviation of up to 37% in the spring constant. Thus, a calibration of the spring constant of the capillary is required after mounting it in the fixture.

For the experimental calibration of spring constant k , we measured the force F using a scale with approximately 1 μ g resolution. We

mounted the capillary fixture to a micromanipulator to be able to move the fixture precisely. Then we put a pin on a microgram scale to ensure that the force that bent the capillary acted only on the free end of the capillary. Using the micromanipulator, s was changed in steps of 50 μm and F was recorded. The measured force versus s plot showed a strong linear correlation (Pearson $R = 0.9999$, see Supporting Information S1). In addition, it can be assumed that there is no constant force offset because there was no force measured when the capillary was not deflected. The experimental calibration showed that Hooke's law is indeed applicable to our system. Fitting the data with a linear function yields $k_{\text{calibration}} = 0.107 \pm 0.003 \text{ N/m}$, which is a factor of 1.38 smaller than the nominal one.

2. Calibration of the LDS Setup. To determine the sensitivity of the LDS, the ratio of the voltage change of the PSD upon deflecting the capillary (S_{LDS}) was measured. Therefore, the free end of the capillary was deflected in steps of 50 μm using a micromanipulator, and the PSD voltage was recorded. The PSD signal changed linearly (Pearson $R = 0.9999$, see Supporting Information S2) with the deflection. When the spring constant was divided by S_{LDS} , the calibration constant $K \approx 10^{-5} \text{ N/V} \pm 5\%$ was determined. Because S_{LDS} depends on the alignment of the LDS components, K needs to be measured after any change in the LDS setup. The force resolution of the DAFI with the above presented components is below 40 nN.

3. Sample Adjustments and Movement. Solid samples were placed on the rim of a circular disk of 5 cm radius. The disk was powered by a linear motor (Faulhaber 2232 A 012 SR with transmission gear 22 EK 23014:1) to ensure a steady motion. The disk was held horizontally and rotated at a constant velocity of 0.015 to 0.15 rpm. The drop and capillary were placed at a radius of 4.8–5 cm, which led to a velocity range of 75–785 $\mu\text{m/s}$. The capillary was aligned relative to the disk so that the normal of the coated surface was parallel to the tangential of the circular disk at that point (i.e., the x direction in Figure 2). Thus, the direction of drop motion was initially parallel to the x axis. The lateral adhesion force needs to act in x direction because this is the direction for which our capillary is calibrated. To ensure an optimal projection of the lateral adhesion force in the x direction, the curvature of the drop trajectory had to be minimized. Thus, the sample was positioned on the rim of the disk. For a moving drop, the lateral adhesion force had approximately a projection of 2% into the y component and a projection of 98% into the x component. This systematic error is considered in the error of K .

4. Imaging System. We used two black and white cameras with a maximal frame rate of 25 fps. The cameras were equipped with manual zoom objectives with a variable focal length from 13 to 130 mm. The cameras were connected to a PC via IEEE-1394a. For data acquisition, we used National Instruments LabVIEW 2011. To acquire the images, the NI-IMAQdx 3.9.1 driver was used. The PSD gave four current signals as output. These were converted to x , y , and sum-voltage signals by a custom-made amplifier. We fed these signals into an NI PCI-6221 DAQ card and used the NI-DAQmx 9.5.1 driver to acquire the signal. The PSD signal was acquired at a sampling rate of 10 kHz. Our capillary had a resonance frequency of about 85 Hz. Our sampling rate was about 10^2 times higher than the highest possible frequency that could be detected by our capillary. Thus, the Nyquist-Shannon sampling theorem was satisfied.

5. Software. The sampling clock for the PSD-voltage data acquisition was derived from the onboard clock of the DAQ card. For synchronization, the cameras were triggered with a trigger signal that was also derived from the onboard clock of the DAQ card. The camera trigger itself was triggered by the first sampling-trigger pulse for the PSD-voltage DAQ, thus avoiding a timing offset. The collected data was saved, including other parameters such as the temperature and humidity (LabVIEW 2011).

6. Preparation of a PDMS Sample. Ethanol (Sigma-Aldrich, 99.8%), hexane (Fisher Chemicals, 95%), sulfuric acid (VWR, 95%), ammonia solution (VWR, 28%), acetone (Sigma-Aldrich, 99.7%), hydrogen peroxide solution (Sigma-Aldrich, 34.5–36.5%), trimethylsiloxy-terminated poly-(dimethylsiloxane) (PDMS) (200 cSt., ABCR), and 24 mm $\times$ 60 mm cover slides (Menzel-Gläser) were used as received. To obtain hydrophobic PDMS surfaces, we followed the

preparation procedure reported by Krumpfer et al.¹⁸ The precleaned cover slides were placed in a 2:1 mixture of sulfuric acid and hydrogen peroxide solution, placed in a water bath, and held for 2 h at 70 $^{\circ}\text{C}$. Then the samples were rinsed with very pure water and ethanol. After being dried, the cover slides were put into 500 mL ultraclean borosilicate glass vials (Duran Pure vials GL45 with a dust cover, Duran Group, Schott) and sufficiently wetted with PDMS. The vials were resealed and placed in an oven for 24 h at 100 $^{\circ}\text{C}$. Then the samples were properly rinsed with hexane, acetone, and Milli-Q water and dried. To characterize the surfaces, we measured the advancing $\theta_a(0)$ and receding $\theta_r(0)$ contact angles with classical sessile drop experiments. The “0” indicates that the contact angles were measured in the limit of $v \rightarrow 0$. These angles were measured on sessile drops by changing the volume of a drop on the solid surface. To measure $\theta_a(0)$, we added water to a sessile drop until the contact line advanced. The addition of water was then stopped, and the contact angle was measured for $\theta_r(0)$. Advancing and receding contact angles for water were $\theta_a(0) = 105 \pm 2^{\circ}$ and $\theta_r(0) = 93 \pm 3^{\circ}$, giving a contact angle hysteresis of $\theta_a(0) - \theta_r(0) = 12^{\circ}$.

7. Preparation of a Periodic Test Structure. First, a pattern of dots (black cartridge, HP Q2670A) was printed on a PVC foil (Avery Zweckform) using a conventional HP office laser printer with a printing resolution of 600 dpi. The patterns of dots were generated using ImageJ. The printed foil was first covered with 10 nm of chromium and then with 100 nm of gold by evaporation. To make the surface hydrophobic, the coated foil was put into a saturated solution of hexadecanethiol in ethanol for 15 min at room temperature. The surface showed contact angles of $\theta_a(0) = 120 \pm 5^{\circ}$ and $\theta_r(0) = 75 \pm 5^{\circ}$ and a roughness of $R_q = 0.872 \mu\text{m}$ over an area of $800 \times 800 \mu\text{m}^2$, based on optical profilometry ($\mu\text{-surf}$). Over a patterned area (i.e., the dots), the contact angles were $\theta_a(0) = 97 \pm 4^{\circ}$ and $\theta_r(0) = 62 \pm 2^{\circ}$. These contact angles were determined on a 1 cm^2 surface that was prepared in the same way that the dots were prepared. The defects had a size of 300–390 μm and an average height of 4 μm , as determined with optical profilometry (μSurf , NanoFocus).

8. Preparation of SU8-Pillar Arrays. Arrays of hydrophobic pillars were prepared by lithography on an SU-8 negative photoresist (MicroChem Corp., Newton, MA). Glass slides with a thickness of 170 μm were cleaned with acetone in an ultrasonic bath and dried in vacuum at 170 $^{\circ}\text{C}$ overnight. An initial layer of SU8 was fabricated, which ensured uniformity and exhibited good adhesion to the glass substrate. First, a 5- μm -thick SU-8 layer (using the SU-8 2005 solution) was spin-coated at 4000 rpm for 30 s on the slides. Second, the coated slides were soft baked at 95 $^{\circ}\text{C}$ for 3 min, cooled slowly, and then exposed everywhere to UV light ($\lambda_{\text{max}} = 360 \text{ nm}$, mercury short arc lamp at 350 W, Advanced Radiation Corporation, Santa Clara, CA) for 25 s. After exposure, the samples were baked at 95 $^{\circ}\text{C}$ for 3 min, allowing the film to cross-link. On this 5 μm film, another 25 μm of SU-8 (SU-8 2025 solution) was spin-coated at 4000 rpm for 30 s. After a soft bake step at 95 $^{\circ}\text{C}$ for 5 min, the samples were exposed to UV light for 35 s with the same lamp through a chromium-coated glass mask having square free areas with spacing $d = 100 \mu\text{m}$, edge $a = 50 \mu\text{m}$, and $h = 23 \mu\text{m}$ (Karl Suss, Garching, Germany, MJB3 mask aligner). Then the samples were baked at 95 $^{\circ}\text{C}$ for 5 min, allowed to cool slowly (overnight), and finally developed with a suitable solvent (MicroChem, Corp., Newton, MA). To hydrolyze the surface of SU-8 and create OH groups, the samples were immersed in 0.1 M NaOH at 25 $^{\circ}\text{C}$ overnight. Finally, the surfaces of the pillars were hydrophobized with (1H,1H,2H,2H)-perfluorooctyl-trichlorosilane. A scanning electron microscope image showed that the side walls are almost vertical (Figure 3). The advancing and receding contact angles of a water drop deposited on these pillars were $\theta_a(0) \geq 162 \pm 3^{\circ}$ and $\theta_r(0) = 120 \pm 10^{\circ}$, and on the flat parts of the substrate, they were $\theta_a(0) = 117 \pm 3^{\circ}$ and $\theta_r(0) = 85 \pm 5^{\circ}$, respectively.

III. RESULTS AND DISCUSSIONS

A. Homogeneous, Smooth Surfaces. Analytical expressions for adhesion forces of drops on homogeneous, smooth surfaces (eq 1) have been derived for tilted-plate experiments.

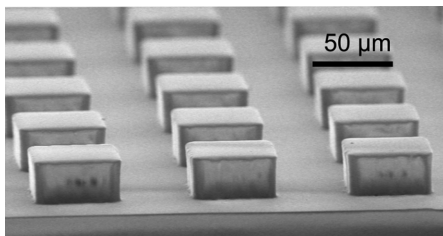


Figure 3. Scanning electron microscope image of SU8 pillars.

Thus, we first investigated a homogeneous, smooth PDMS surfaces. We can write

$$F_{LA}(v) = \frac{24}{\pi^3} \gamma D (\cos \theta_r(v) - \cos \theta_a(v)) \quad (3)$$

which is similar to eq 1. Here we assume that the TPCL of a drop remains elliptical during dynamic measurements. We further assume that the contact angle of a sliding drop varies along the TPCL in the same way as for drops on the verge of sliding. Also, the capillary that was more than 0.5 mm away from the TPCL should not influence the dependence of the contact angle on the position along the TPCL.¹⁹ Equation 3 relates the dynamic lateral adhesion force $F_{LA}(v)$ to the shape of the drop. θ_{min} and θ_{max} in eq 1 were substituted with $\theta_r(v)$ and $\theta_a(v)$, the dynamic advancing and receding contact angles, because the drop is in motion when $F_{LA}(v)$ is measured. In the case of moving drops, the contact angles are independent of the initial conditions.¹⁵ Unlike θ_{max} and θ_{min} , the dynamic contact angles depend on the velocity: $\theta_a(v)$ increases with increasing velocity whereas $\theta_r(v)$ decreases.^{20–22}

We investigated the size and velocity dependencies of the measured $F_{LA}(v)$ on smooth, homogeneous PDMS and compared the measured values with calculated forces using eq 3. To investigate the D dependency, we deposited drops of 2 μL volume and let them evaporate to decrease their size. From the drop profiles measured with the cameras, the average width and length of the contact area were derived. From these we calculated D . To investigate the velocity dependence, we moved a drop over the same trajectory at different velocities.

The result of a single measurement, the force measured with the capillary versus time, is plotted in Figure 4. The plotted data was recorded with a water drop of 1 μL volume on PDMS at a velocity of $v = 251 \pm 10 \mu\text{m/s}$. Five characteristic regions and one characteristic event are observable. The drop was positioned close to the capillary but was not yet in contact (region 1). The measured force was zero. The sample surface was then moved relative to the capillary to the right. When the capillary first touched the drop, it was instantaneously wetted by the liquid and bent toward the drop. We call this event “snap-in”. The measured force was negative. In the second region, the surface moved further until the capillary passed the drop apex. A force corresponding to the peak in region 2 was necessary so that the capillary could penetrate the drop. In the third region, the capillary approached the rim of the drop surface. In region 4, the capillary started to pull on the drop. The static lateral adhesion force was still stronger than the force exerted by the capillary, so that the drop was not yet in motion relative to the surface. Eventually, the drop started to slide (region 5), and the drop moved relative to the surface.

In region 5, the dynamic lateral adhesion force was measured. To move the drop, the adhesion between the capillary and the drop had to be stronger than the lateral adhesion between the

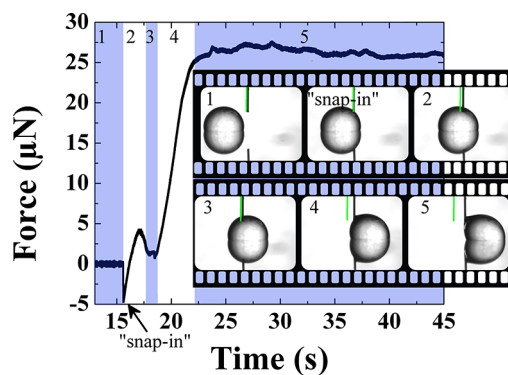


Figure 4. Measurement of the lateral adhesion force versus time on a PDMS sample. The experiment was performed at $v = 251 \mu\text{m/s}$ with a water drop of volume 1 μL . The recorded force signal is divided into five regions, marked alternately with blue and white. The filmstrips show snapshots recorded with the orthogonal camera (7 in Figure 2) corresponding to the five regions and the snap-in event. Below the actual drop, the drop's reflection is visible. In the middle of each snapshot one can see the capillary (black line). The green line in the snapshots indicates the position of an undisturbed capillary.

drop and the surface. We were always able to move the drop for $\theta_a \geq \theta_r > 80^\circ$. For $\theta_r < 45^\circ$, the drop adhered so strongly to the surface that the capillary was not able to slide it over the surface. Once the drop was sliding, the average force (region 5) was $26.3 \pm 2.6 \mu\text{N}$. In addition, the signal showed weak variations of the force in the range of 25.4 to 27.6 μN that can be attributed to the stick–slip movement of the TPCL. For the analysis in further experiments, only region 5 was considered because only this region contains information about the dynamic lateral adhesion force at constant velocity. The tangent of the drop at the TPCL gives the contact angles that depend on the velocity: $\theta_a(v) = 113 \pm 3^\circ$ and $\theta_r(v) = 93 \pm 1^\circ$ for $v = 251 \mu\text{m/s}$. With $D = 1.29 \pm 0.04 \text{ mm}$, eq 3 leads to $F_{LA}(v) = 24.3 \pm 3.7 \mu\text{N}$, which is in good agreement with the measured force.

1. Size Dependence of F_{LA} . To quantify the dependence of $F_{LA}(v)$ on the drop size, seven measurements were carried out with a drop of approximately 2 μL initial volume. For every measurement, the dynamic lateral adhesion force and D were recorded on a trajectory of 2.5 mm length (10 s) and then averaged. Between measurements, we waited several minutes so that the water partially evaporated and the drop decreased in size. F_{LA} increased linearly with increasing D (Figure 5). A linear function with the intercept at zero could be fitted to the data points (solid line in Figure 5). The intercept was set to zero because by definition the force needs to be zero in the absence of a drop. The Pearson R for the data points was 0.9999, thus proving a strong linear correlation of the data, as expected from eq 3. The slope of the linear fit function was $18.3 \pm 0.9 \mu\text{N/mm}$.

To compare this value to previous findings, we calculated $F_{LA}(v)/D$ using eq 3. We have assumed that the contact angles were the same for all seven measurements because the measurements were performed on the same trajectory at the same velocity. We have measured $\theta_r(v)$ and $\theta_a(v)$ optically for the first five of the seven measurements. In the last two measurements the drop was too small for providing good enough contrast for measuring the contact angles. Averaging gave $\theta_r(v) = 92 \pm 2^\circ$ and $\theta_a(v) = 113 \pm 2^\circ$, which led to

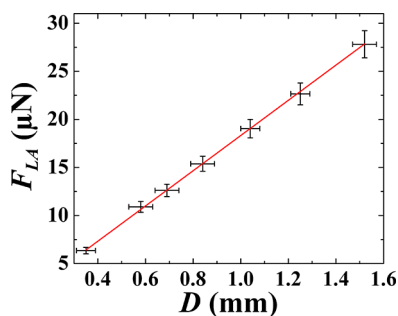


Figure 5. Lateral adhesion force $F_{LA}(v)$ versus D measured on PDMS at $v = 251 \mu\text{m/s}$. Force values were averaged over 10 s, corresponding to a sliding distance of 2.5 mm. Error bars for $F_{LA}(v)$ correspond to spatial force variations and the 5% calibration error. In our measurements, the relative humidity was about 50%. D declined by roughly 0.05 mm/min. The error bars for D correspond to drop evaporation during each measurement and a reading error. The red line is the linear fit.

$F_{LA}(v)/D = 19.9 \pm 2.6 \mu\text{N/mm}$. This value is in good quantitative agreement with the measured slope.

We like to emphasize that for drops with volume of $<2 \mu\text{L}$ tilted plate experiments could not be performed on the PDMS surface. Even for high tilting angles the lateral adhesion force exceeds the gravitational force acting on the drop.

This linear relationship allows us to deal with one problem encountered when studying water: drop evaporation. For long measurements and to compare measurements with different drops better, we report $F_{LA}(v)/D$ instead of $F_{LA}(v)$ because the quotient is independent of the drop volume and thus not affected by the evaporation of the liquid.

2. Velocity Dependence of F_{LA} . The velocity dependence was studied in the range of $v = 75 - 753 \mu\text{m/s}$ using drops of roughly $2 \mu\text{L}$ volume. The measured force increased linearly with increasing velocity (Figure 6). The reason is that $\theta_r(v)$

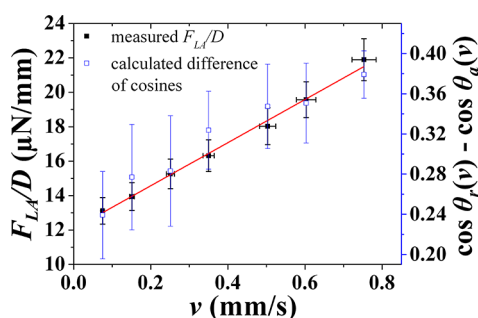


Figure 6. F_{LA}/D vs velocity measured on PDMS (filled black squares). F_{LA}/D was plotted instead of F_{LA} to account for drop evaporation. The drop volume was approximately $2 \mu\text{L}$. Force values were averaged over a sliding distance of 0.6 mm. The results have been fitted with a linear function according to molecular kinetic theory (red line). Calculated differences in the cosines of measured contact angles (open blue squares) show the same trend as the measured F_{LA}/D .

decreases and $\theta_a(v)$ increases with increasing velocity. As a result, $\cos \theta_r(v)$ increases and $\cos \theta_a(v)$ decreases so that the difference, $\cos \theta_r(v) - \cos \theta_a(v)$, increases (eq 3).

The velocity dependency of the dynamic contact angles can be fitted by molecular kinetic (MK) theory.^{21,22} We chose MK theory because for large contact angles and low velocities the dynamics are more likely to be controlled by contact line friction.^{23,24} Contact line friction may be represented by MK

theory. According to MK theory, the velocity and the dynamic contact angle at a moving TPCL are related by

$$v = 2K_0\lambda \sinh\left[\frac{\lambda^2\gamma}{2k_B T}(\cos \theta(0) - \cos \theta(v))\right] \quad (4)$$

Here, K_0 is the equilibrium frequency of the random displacement of water molecules at the TPCL, λ is the distance between adsorption sites, $\theta(0)$ is the contact angle for $v \rightarrow 0$, and $\theta(v)$ is the dynamic contact angle. For a TPCL that has stopped advancing, $\theta(0)$ approaches $\theta_a(0)$ ($\theta_a(0) = \lim_{v \rightarrow 0} \theta_a(0)$) and $\theta_r(0) = \lim_{v \rightarrow 0} \theta_r(v)$. For the low velocities at which DAFI operates, the argument of the hyperbolic sine is small. Thus, eq 4 can be simplified to

$$\begin{aligned} v &= \frac{K_0\lambda^3\gamma}{k_B T}(\cos \theta(0) - \cos \theta(v)) \\ &\equiv \frac{\gamma}{\zeta}(\cos \theta(0) - \cos \theta(v)) \end{aligned} \quad (5)$$

with the friction coefficient $\zeta \equiv k_B T/(K_0\lambda^3)$. According to eq 5, the velocity of the advancing contact line is given by

$$\begin{aligned} v &= \frac{\gamma}{\zeta}[\cos \theta_a(0) - \cos \theta_a(v)] \\ \Rightarrow \cos \theta_a(v) &= \cos \theta_a(0) - \frac{v\zeta}{\gamma} \end{aligned} \quad (6)$$

and the velocity of the receding line (that is moving in the opposite direction) is given by

$$\begin{aligned} -v &= \frac{\gamma}{\zeta}[\cos \theta_r(0) - \cos \theta_r(v)] \\ \Rightarrow \cos \theta_r(v) &= \cos \theta_r(0) + \frac{v\zeta}{\gamma} \end{aligned} \quad (7)$$

Inserting $\cos \theta_a(v)$ and $\cos \theta_r(v)$ into eq 3 leads to

$$\frac{F_{LA}(v)}{D} = \frac{48}{\pi^3}\zeta v + \frac{24}{\pi^3}\gamma(\cos \theta_r(0) - \cos \theta_a(0)) \quad (8)$$

The dynamic lateral adhesion force is thus expected to increase linearly in v . The measured velocity dependence was linear with a Pearson R of 0.998 and is thus in good agreement with MK theory (Figure 6). We have fitted the data with a linear function and obtained a slope of $12.5 \pm 0.6 \mu\text{Ns/mm}^2$ (i.e., a friction coefficient of $\zeta = 8.1 \pm 0.4 \mu\text{Ns/mm}^2$) and an intercept of $12.1 \pm 0.6 \mu\text{N/mm}$. The intercept corresponds to the lateral adhesion force for $v \rightarrow 0$, which is a quantitative measure of the static contact angle hysteresis. The intercept was in good agreement with $\theta_a(0) = 105 \pm 2^\circ$ and $\theta_r(0) = 93^\circ \pm 3^\circ$ measured with the sessile drop method, which when inserted into eq 8 (for $v = 0$) gave $F_{LA}(0)/D = (11.5 \pm 3.5) \mu\text{N/mm}$. Thus, we can successfully fit the measured force with MKT theory. Whether the derived friction coefficient corresponds to actual microscopic jump events at the TPCL remains speculation.²¹

B. Pinning at Single Defects. On anisotropic and inhomogeneous surfaces, the TPCL can no longer be approximated with an ellipse because single defects can strongly deform the TPCL.²⁵ Consequently, eq 3 is no longer applicable to these surfaces. In addition, most optical arrangements are focused on the leading and trailing edge of the drop in order to detect θ_a and θ_r . Thus, the noncentral pinning of the TPCL (i.e., pinning that happens distant from

the leading or trailing part of the TPCL) cannot be detected with such optical arrangements. Only the pinning of the TPCL and the force due to these pinning can be detected when the complete shape of the drop is monitored. In principle, this is possible with the image-based adhesion force analysis (IBAF) method³ or with confocal microscopy;¹⁹ however, these methods are typically too slow to follow the dynamics of the TPCL at the velocities considered here.

This aspect (i.e., the measurement of forces owing to the pinning of the TPCL to single defects) can be investigated by the DAFI elegantly. We have fabricated a periodic test structure where pinning of the drops at defects occurred at different positions on the TPCL (Figure 7b). This pinning at different

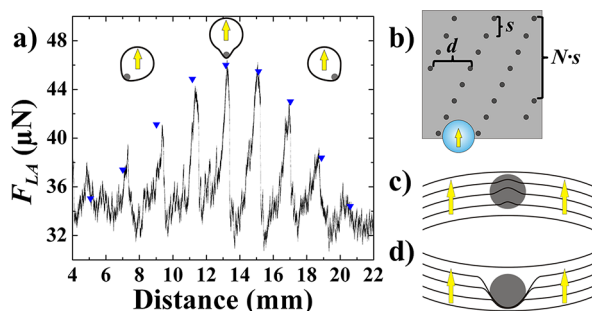


Figure 7. (a) Lateral adhesion force of a moving water drop on a periodic test structure at $v = 151 \mu\text{m/s}$. The blue triangles are the theoretically estimated peak positions and heights (Supporting Information S3). The deformed black circles and gray dots sketch the deformation of the TPCL due to the pinning of the drop at a defect. (b) Layout of the test structure. The surface is hydrophobic and has a structure of periodically ordered uniform defects that are less hydrophobic. d and s are the distances between defects orthogonal and parallel to the direction of drop motion, respectively. $N \cdot s$ is the period of the structure. A drop is moved in the direction of the arrow over the surface. (c) Black lines are consecutive configurations of the TPCL when moving over a defect (gray disk). The yellow arrows indicate the direction of motion of the TPCL. The advancing TPCL gradually moves over a defect. (d) The receding TPCL pins to and then jumps over a defect.

positions on the TPCL leads to position-dependent pinning forces. A defect that pins the TPCL at the very end of the trailing edge of the drop leads to the highest possible pinning force because the projection of the surface tension in the direction of motion of the drop is maximal. A defect that pins the TPCL more to the left or right, relative to the direction of motion, leads to a weaker pinning force.

When we use the DAFI, the movement of the drop is not perfectly straight over a surface but has a circular trajectory with a radius of roughly 5 cm. The pattern shown in Figure 7b was adjusted to account for the circular trace of the drop. Here, d and s were set to be 2 mm, thus larger than the size of the drop length in the direction of motion. This spacing guaranteed that the TPCL was in contact with only one defect at a time. N was set to 10 (i.e., in the direction orthogonal to the direction of drop motion, the dots were shifted mutually by 0.2 mm). While the drop moved over the test structure, the orthogonal projection of the solid–liquid interface showed a width of 1.03 mm and the parallel projection showed a width of 0.8 mm.

Figure 7a shows the measured force when the drop was moved over one period of our test structure. The nine peaks correspond to the pinning of the TPCL to nine defects. As expected, the peaks had different heights because the pinning

force depends on the position of pinning on the TPCL (deformed black circles with gray dots).

The TPCL moved over every defect twice: first with the advancing part and afterwards with the receding part of the TPCL. Thus, one could expect two force peaks for every defect. The positive force peaks in Figure 7a correspond to the pinning of the receding part of the TPCL. The receding contact angle on the defects ($\theta_r(0) = 62 \pm 2^\circ$) was smaller than the receding contact angle on the rest of the surface ($\theta_r(0) = 75 \pm 5^\circ$), which impeded the recession of the TPCL.²⁶ The receding part of the TPCL pinned to the defect until the receding contact angle of the defect was reached. Then the TPCL abruptly depinned (Figure 7d). The advancing contact angle on the defects ($\theta_a(0) = 97 \pm 4^\circ$) was also lower than the advancing contact angle on the rest of the surface ($\theta_a(0) = 120 \pm 5^\circ$). Thus, the advancing TPCL should have jumped ahead over the defects. This would have led to a negative force peak. Because the part of the TPCL that was in contact with the defect was confined by parts of the TPCL that were in contact with the surrounding surface, the jumping was hindered and the part of the contact line that was in contact with the defect advanced gradually (Figure 7c). Thus, the effect of the defects on the advancing contact line was insignificant compared to the effect on the receding contact line.

From knowing the relative positions of the defects and approximating the TPCL with a circle, we estimated the position of the force peaks caused by the pinning of the TPCL to the defects (i.e., the x coordinate of the blue triangles in Figure 7a (Supporting Information S3)). To estimate the relative height of the force peaks, we assumed that the additional force due to pinning to a defect is proportional to the difference in the projections of the surface tension in the direction of drop motion. The y coordinate of the blue triangles gives an estimate of the peak heights (Figure 7a; Supporting Information S3). The peak heights agree nicely with our calculation. Thus, the measured peaks of the force signal indeed correspond to the defects on the periodic test structure.

C. Artificial Superhydrophobic Pillar Structure. The minimization of adhesion forces is essential for superhydrophobic (i.e., water-repellent) surfaces. Superhydrophobicity is due to surface roughness that allows air cushions to be trapped between the water and the substrate. The above measurements show that the DAFI can be used to measure drop adhesion on surfaces with defects; therefore, it is an ideal tool for studying superhydrophobicity. Here we apply the DAFI in order to understand the wetting of artificial pillar arrays. Artificial pillar arrays were often used as model systems to deepen the understanding of superhydrophobicity and contact angle hysteresis.^{5,27–29} If a drop is placed on a superhydrophobic surface, then it forms a contact angle with the surface that exceeds 150° and the drop rolls off when tilting the surface by less than 10° . Compared to that in a flat solid, surface adhesion is strongly reduced in the superhydrophobic state because the drop mainly rests on air cushions.^{5,18,30–33} The solid–liquid interface is restricted to the pillar's top. After a $2 \mu\text{L}$ water drop was positioned on the SU8 pillars, the air cushion separating the substrate and the drop could be seen. This means that the drop was in the Cassie state. We moved the drop over the sample as sketched in Figure 8a.

Our measurements revealed a mean lateral adhesion force of $12.8 \mu\text{N}$ (Figure 8b). Spatial variations of the force in the form of peaks of apparently $2.8 \mu\text{N}$ peak-to-peak height with a mean spacing of $101 \pm 9 \mu\text{m}$ were observed. This distance

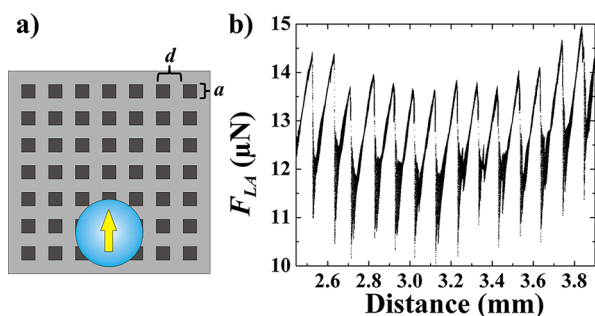


Figure 8. (a) Sketch of the SU8 pillar array. $d = 100 \mu\text{m}$, $a = 50 \mu\text{m}$, and $h = 23 \mu\text{m}$. The drop is blue, and the pillars are dark gray (not to scale). The yellow arrow shows the direction of motion of the drop. (b) Force variation recorded when moving a $2 \mu\text{L}$ water drop over the SU8 pillar array at $v = 75 \mu\text{m/s}$.

corresponds to the period $d = 100 \mu\text{m}$ of the structure (Figure 8a). The overall first decrease and subsequent increase in the lateral adhesion force are most likely effects of the circular trajectory of the drop (Figure 8b).

The dynamic advancing contact angle was $\theta_a(75 \mu\text{m/s}) \geq 165^\circ$. The method of determining the contact angles from the projection of the drop shape has a limitation that is due to the perspective at which the camera looks at the drop. Contact angles close to 180° cannot be resolved properly. Thus, we could only give a lower limit for $\theta_a(v)$. Because of the high dynamic advancing contact angle, we assume that the leading part of the drop advanced by touching and wetting the next row of pillars.^{18,34} Note that the contact angles are apparent macroscopic contact angles corresponding to the Cassie state, measured on a length scale that is much longer than the microscopic surface structure. On superhydrophobic surfaces, the microscopic or material contact angle, obtained from extrapolating the liquid surface down to a distance of 10–100 nm, may be very different.¹⁹

To move from one pillar to the next, the receding drop rim had to depin from one pillar and jump to the next (Figure 9).³⁵

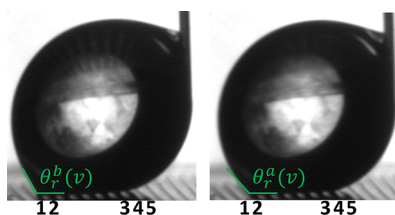


Figure 9. Shape of the drop when proceeding over the pillar structure. The left picture was taken just before a transition, and the right, just after a transition. The time between pictures is 40 ms. The black numbers label adjacent rows of pillars that are either in contact with the advancing or receding part of the drop rim. The receding drop rim jumps from row 1 (left) to row 2 (right). The advancing drop rim wets row 3 but not row 4 in the left picture and row 4 but not row 5 in the right picture.

The rim has to transition from the so-called pinning mode to the depinning mode.³⁶ Such depinning–pinning transitions were recently found to influence the energy barrier needed for Cassie–Wenzel wetting transitions.³⁷ Receding depinning–pinning transitions from one pillar to the next required a relative force that we will call F_{trans} . Receding and advancing events appeared synchronous in the video taken with the

orthogonally adjusted camera. (See snapshots of the video in Figure 9.) We thus have to assume that both receding depinning and advancing wetting events contributed to each force peak in Figure 8b. $\theta_a(v)$ exceeded 160° and did not change significantly. $\theta_r(v)$ changed from $\theta_r^b(v) = 118 \pm 2^\circ$ just before depinning to $\theta_r^a(v) = 127 \pm 2^\circ$ directly after depinning; the additional indices indicate before “b” and after “a” depinning. Because the cosine function is weakly dependent on the angle θ_a , when θ_a is close to 180° the contribution of the advancing contact line to the measured force peaks was rather insignificant, in accordance with refs 26 and 30. The drop width orthogonal to the direction of motion was $0.50 \pm 0.03 \text{ mm}$, which corresponds to five pillars that were in contact with the receding part of the drop rim (Supporting Information S4). If the receding part of the drop rim would have depinned from one pillar at a time,³³ then additional peaks in the force curve less than $100 \mu\text{m}$ apart would have shown up in Figure 8b. From the correspondence of the force peak distance and the period d of the structure we conclude that most depinning–pinning transitions were concerted. Thus from analyzing the peak height and determining the number of pillars involved in a transition the relative depinning–pinning transition force per pillar F_{trans} can be calculated.

After every peak in Figure 8b, the force oscillated at a frequency of $51 \pm 1 \text{ Hz}$. The sudden depinning corresponds to the broadband excitation of the capillary. This vibration frequency can be explained by the resonance frequency of the capillary being damped by the drop. The resonance frequency of a free liquid drop is about $f = (2\gamma/\pi^2 R^2 \rho)^{1/2} \approx 230 \text{ Hz}$. Calculating the resonance frequency for a sessile drop with a contact angle of 150° leads to a slightly higher resonance frequency compared to a free liquid drop.^{38–40} This resonance frequency of the drop is much higher than the observed 51 Hz.

The mean value of the force oscillation corresponds to the force actually measured after a depinning–pinning transition occurred. We have thus determined F_{trans} by calculating the difference between the force just before the depinning occurred and this mean value after depinning. We have determined the transition force of the complete receding part of the drop rim to be $2.27 \pm 0.62 \mu\text{N}$. The error is the standard deviation plus an additional error of $0.5 \mu\text{N}$ due to an increased oscillation just before depinning and the 5% error in calibration. Dividing this value by the number of pillars with which the receding part of the drop rim was in contact (i.e., 5) leads to a transition force per pillar of $F_{\text{trans}} = 0.45 \pm 0.15 \mu\text{N}$.

To estimate the relative depinning–pinning transition force, we took the difference in the projections of the surface tension in the direction of motion before and after the depinning–pinning transition times the length of the contact line:

$$F_{\text{trans}} = d\gamma(\cos \theta_r^b(v) - \cos \theta_r^a(v)) \quad (9)$$

Each pillar has to bear the force of a length d of the rim of the drop. This led to $F_{\text{trans}} = 0.96 \pm 0.30 \mu\text{N}$. We believe that the difference between this value and the measured is due to the curved shape of the drop rim.

IV. SUMMARY AND CONCLUSIONS

With the DAFI, the lateral adhesion force of liquid drops on solid surfaces can be measured. We demonstrated that the measured lateral adhesion forces could be related to the shape of the drop using eq 3. The lateral adhesion force was found to be linearly dependent on the velocity. Most other methods

applied in this field provide only information about static lateral adhesion forces or the roll-off angle.^{11,41} In particular with the DAFI, $F_{LA}(v)$ could be measured spatially resolved. Thus, structured surfaces and gradients in surface tension can be quantified.^{16,42–44} Furthermore, variations of the force caused by pinning of the TPCL to single defects were resolved. Single defects that are more wettable than their surrounding surface were found to affect drop movement mainly by pinning the receding part of the TPCL. The lateral adhesion force of a superhydrophobic SU-8 pillar-array showed constant and spatially dependent domains. The spatially dependent contributions were found to originate mainly from the depinning–pinning transitions of the receding part of the drop rim, and the relative depinning–pinning transition force per pillar was determined.

Our measurements show that static quantities such as static contact angles or sliding angles are insufficient to characterize the lateral adhesion force fully. Using the DAFI, we were able to determine both dynamic and static lateral adhesion forces, confirming that the DAFI is a suitable tool for characterizing lateral adhesion and thus hydrophobicity. In particular, the relative movement of the drop on surfaces allows us to resolve different wetting phenomena of surfaces spatially. Surface areas of up to 1 cm² can be investigated within 1 min. Thus, the DAFI could even be applied for quality control of surfaces made by large-scale industrial processes.

■ ASSOCIATED CONTENT

■ Supporting Information

Experimental calibration of a capillary. Calibration of the sensitivity of the laser deflection system. Calculation of peak position and height in Figure 7b. Determining the number of pillars in contact with the receding drop rim. This material is available free of charge via the Internet at <http://pubs.acs.org>

■ AUTHOR INFORMATION

Notes

The authors declare no competing financial interest.

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